

Week 10: Relefection

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Most Challenging Aspect and How I Overcame It

The most challenging part of this course was returning to Python after a long break while learning unfamiliar tools at the same time. I had not written Python in a while, so the basic syntax took relearning, and on top of that I had to learn pandas operations like pivoting long-format data into wide format and merging multiple data sets. The visualization libraries added a third layer of difficulty, because I had to learn not just how to write the code but how to choose the right chart for the question I was trying to answer. I overcame this by breaking the work into small steps and treating each error as a learning moment rather than a setback.

Applying a Data Literacy Skill in the Future

The specific data literacy skill I plan to apply going forward is designing visualizations for distinct audiences. In the Week Nine project I built one chart for corporate executives and a different chart for county commissioners, even though both drew from the same underlying data set. I plan to carry this skill into a personal project I am developing, a 3D business directory aimed at small businesses without strong digital presences, similar in concept to platforms like Yelp. That platform will have two audiences: business owners who need to manage their listings and consumers who need a clean way to discover local options. Each audience requires a different visual treatment of the same underlying data, and I now know how to think about that distinction explicitly.

How Visualizations Supported the Data-Driven Narrative

In the Week Nine project, my visualizations directly supported a data-driven narrative about uneven workforce development across the United States since 2020. The high-level bar chart compressed a composite Expansion Attractiveness Score into a scannable ranking of the

top fifteen states, with color encoding median household income as a second attribute. The detailed bubble scatter then expanded the same data into a four-dimensional view, plotting educational attainment against unemployment, with bubble size for population and color for population growth. Together, the two visualizations carried the story from a strategic question for executives to a diagnostic resource for commissioners. Without them, the narrative would have been a list of numbers; with them, the reader could see the regional divergence the data described.

How My Perspective on Data Has Changed

My approach to data has shifted dramatically over the course of this semester. I entered the class thinking data analysis began and ended with accurate numbers and polished charts. What I discovered is that the storytelling layer is just as vital as the analysis itself; even the most accurate data provides zero value if it fails to connect with the audience. Moving forward, I view data work as a collaborative conversation between the data, the analyst, and the stakeholders, with compelling narratives and targeted visualizations acting as the bridge that makes the analysis truly actionable.